Detecting Network Scanning Activity Using Nmap and Wireshark

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1. Background and Motivation

Research Problem and Importance

Network scanning identifies open ports and services on a device. Administrators use it to check systems, but attackers can also use scans to gather information before attempting an intrusion. The project asks whether a defender can recognize common Nmap scan behavior by examining captured TCP traffic.

The goal is to find a small set of understandable packet indicators that distinguish controlled scans from ordinary connections in a lab. The work is an experiment and analysis, not a claim that all scans can be detected.

Challenges

Legitimate programs also open several network connections, so a detector could mistake normal activity for scanning. Different scan types produce different TCP patterns, and open or closed ports change the responses. A small virtual lab will not represent every real network.

Objectives (Exploration Space)

Compare normal TCP activity with Nmap TCP connect scans and SYN scans. Identify useful indicators such as new destination ports, SYN patterns, connection completion, resets, and timing. Try simple detection rules and report how often they flag scans and normal traffic.

2. Proposed Methodology

Possible Solutions (Ideas)

Use a controlled virtual network with a scanning machine and a target that the researcher owns or is authorized to test. Capture traffic in Wireshark. First generate normal connections, then run Nmap TCP connect (-sT) and SYN (-sS) scans. Save the packet captures and the Nmap command/output for comparison.

Start with simple rule ideas: many distinct destination ports contacted in a short time, repeated SYN attempts, and patterns of incomplete versus completed TCP handshakes. Adjust the rules based on observed traffic, not assumptions.

Evaluation

Implementation and experimental platform: Nmap, Wireshark, and isolated virtual machines; an optional small script or spreadsheet can count packets or connections. Run each scan type more than once, keep the settings recorded, and repeat baseline traffic captures.

For each known scan and baseline capture, record whether a rule triggers. Report detections, misses, false positives, and a short comparison of TCP connect versus SYN behavior. Use simple descriptive analysis; a mathematical detection model or full intrusion-detection product is not required for the proposed experiment.

3. Expected Contributions

Outcomes

Packet captures and a summary comparing baseline traffic, TCP connect scans, and SYN scans; a table of traffic indicators and observed limitations.

Contributions

A small, repeatable experiment showing which straightforward TCP indicators are useful and when they can produce false positives. The contribution is the documented comparison and evaluation, rather than an entirely new detection algorithm.

Impacts

The findings could help beginners understand what network scanning looks like in packet captures and why alert rules must be checked against legitimate traffic.

4. Targeted Conference/Journal

Venue and Justification

A relevant research venue is the International Symposium on Research in Attacks, Intrusions and Defenses (RAID). It covers intrusion detection, empirical measurement, and evaluation of defensive techniques. The project's controlled packet analysis is related to that scope; naming this venue is for the class template and does not mean a paper will be submitted. See the conference scope at https://raid2026.org/call.html.

5. Organization of the Project and Project Management

5.1 Research Team

Researcher: Alexandro Segura (solo project). Department: Computer Science and Engineering, Oakland University. Email: [ADD YOUR OU EMAIL BEFORE SUBMITTING].

Short bio / strengths / weaknesses: [EDIT THIS WITH YOUR ACTUAL EXPERIENCE]. Possible starting point: "I have worked with basic networking and security topics in class. I will handle the lab setup, captures, analysis, and writing myself. I still need more experience evaluating detection accuracy, so I will keep the experiment small and repeatable."

Because the scope is limited to two Nmap scan types and one isolated environment, the work is planned to be manageable by one student.

5.2 Research Plan

September: choose the topic, prepare the proposal, and make a project website. October: research TCP scanning, design the experiment, and submit the first status report by October 31. November: set up the test lab, capture normal and scan traffic, compare patterns, and submit the second status report by November 14. December: refine and evaluate rules, give the December 12 presentation, and submit the final paper by 1:00 PM on December 13.

Month	Research activities
Sep	Proposal and website
Oct	Research, method design, report 1 (Oct 31)
Nov	Lab setup, capture/analysis, report 2 (Nov 14)
Dec	Evaluation, presentation (Dec 12), final paper (Dec 13)

5.3 Project Management

All project tasks—research, lab setup, testing, analysis, writing, and website updates—are assigned to the solo researcher. Review progress each week, keep copies of commands and captures, and consult the instructor if testing or scope problems arise.

References

Related groups and papers for background research. These are not project collaborators. The sources below are starting points for reading, not a claim that every paper has been reviewed in full.

University Research Groups

1. UC San Diego — CAIDA. Internet traffic measurement and analysis; useful background for interpreting captured traffic. https://www.caida.org/about/

2. Carnegie Mellon — CyLab. Security research including threats and network defense. https://www.cylab.cmu.edu/research/

3. MIT CSAIL — Networks and Mobile Systems. Network protocols, network security, and networked systems. https://www.csail.mit.edu/research/networks-and-mobile-systems

4. Purdue — CERIAS. Network security and intrusion prevention/detection research. https://www.cerias.purdue.edu/

5. University of Michigan — SPQR. Security and privacy research on connected systems; a broader related group. https://web.eecs.umich.edu/~kevinfu/

Company Research Groups

1. Microsoft Research — Security Research Group. Develops and studies security approaches for real-world systems. https://www.microsoft.com/en-us/research/group/security/

2. Cisco Talos. Threat intelligence and detection research for defending networks. https://www.cisco.com/site/us/en/products/security/talos/index.html

3. Palo Alto Networks — Unit 42. Threat research and incident investigation; relevant to detecting suspicious activity. https://www.paloaltonetworks.com/unit42/about

4. IBM — X-Force. Threat intelligence research and security response. https://www.ibm.com/x-force

5. Google — Project Zero. Vulnerability research; broader security context rather than a port-scan-specific group. https://projectzero.google/about-pz.html

Research Papers

1. Jaeyeon Jung, Vern Paxson, Arthur W. Berger, and Hari Balakrishnan (2004). “Fast Portscan Detection Using Sequential Hypothesis Testing.” IEEE Symposium on Security and Privacy.. Directly relevant to how scan activity can be detected from connection patterns. https://people.csail.mit.edu/awberger/papers/fastPortscanDetectionUsingSequentialHypothesisTesting_JungPaxsonBergerBalakrishnan_2004.pdf

2. Vern Paxson (1998). “Bro: A System for Detecting Network Intruders in Real-Time.” 7th USENIX Security Symposium.. Shows how network traffic can be passively monitored and interpreted for intrusion detection. https://www.usenix.org/conference/7th-usenix-security-symposium/bro-system-detecting-network-intruders-real-time

3. Martin Roesch (1999). “Snort: Lightweight Intrusion Detection for Networks.” LISA ’99.. Describes packet-based detection and rules for suspicious traffic. https://static.usenix.org/events/lisa99/full_papers/roesch/roesch_html/index.html

4. Zakir Durumeric, Eric Wustrow, and J. Alex Halderman (2013). “ZMap: Fast Internet-wide Scanning and Its Security Applications.” 22nd USENIX Security Symposium.. Provides background on how scanning tools send probes and why scanning has both defensive and offensive uses. https://www.usenix.org/conference/usenixsecurity13/technical-sessions/paper/durumeric

5. Markus Ring, Sarah Wunderlich, Deniz Scheuring, Dieter Landes, and Andreas Hotho (2019). “A Survey of Network-based Intrusion Detection Data Sets.” Computers & Security, 86, 147–167.. Useful for thinking about what makes baseline and test data suitable for evaluation. https://www.sciencedirect.com/science/article/pii/S016740481930118X

DRAFT: Fill in your OU email and edit the short bio/strengths/weaknesses before submission. Confirm this text matches your approved proposal and actual progress.